



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal to S.S.M.FUEL SERVICE IOC
DEALERS-KSK**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

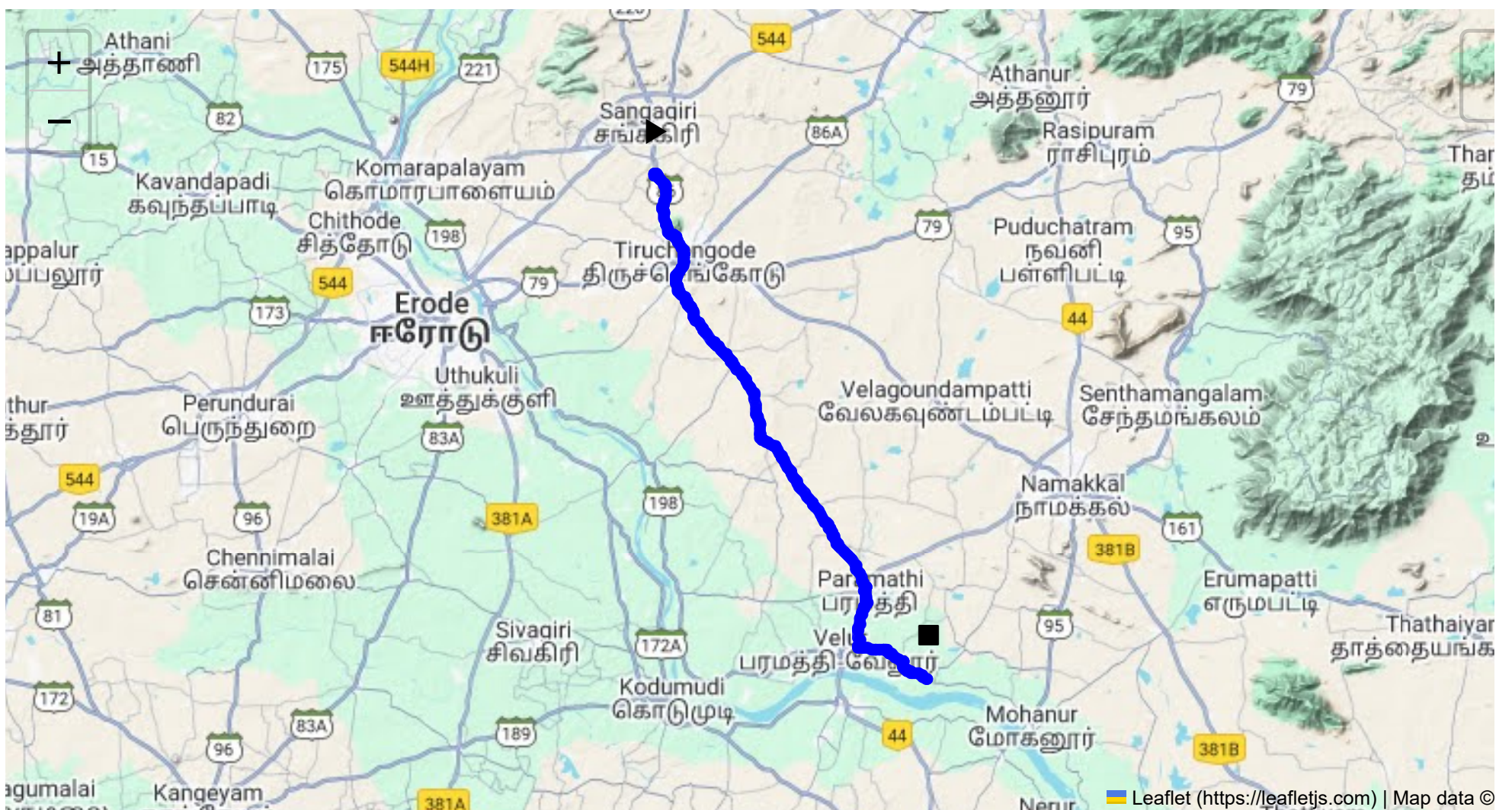
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 48.88 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.092485, 78.06411)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 33V7+2M6, Palapatti spans approximately 48.88 kilometers through rural and semi-urban areas in Tamil Nadu, India. It primarily utilizes a combination of local roads and state highways, ensuring a mix of single-lane and multi-lane roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. This region often faces heavy rains during the monsoon season (October to December), which can lead to flooding and waterlogging. In the summer months (March to May), high temperatures can affect vehicle performance.

3. Analysis of Traffic Patterns

Traffic on local roads varies; mornings (8 AM - 10 AM) and evenings (5 PM - 7 PM) typically see higher congestion due to local commuting and school timings. Traffic is relatively lighter in the early afternoon and late evening, reducing congestion.

4. Assessment of Road Quality and Infrastructure

The route includes well-maintained state highways; however, local roads might exhibit wear and tear, with potential potholes affecting driving conditions. Road signage might be inconsistent, particularly on smaller rural roads.

5. Suggestions for Alternative Routes for Emergencies

- **NH 44:** As a national highway, it offers better infrastructure and emergency services. Using NH 44, connected by nearby entry ramps, provides an efficient alternative during emergencies like roadblocks.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows strict regulations on transporting hazardous materials. It's mandatory to have proper documentation and adhere to specific time windows, generally avoiding densely populated areas during peak hours.

7. Overview of Historical Incidents

There have been few reported incidents involving hazardous materials in this region. However, the most common concerns involve spills and breakdowns due to road conditions during monsoons.

8. Environmental Considerations and Sensitive Areas

The route crosses agricultural regions where spillage or pollution could damage crops. The route must avoid sensitive ecological zones like water bodies to minimize environmental impact in case of incidents.

9. Analysis of Communication Coverage

Most of the route has good communication coverage, but some rural and semi-urban sections might experience weak signals or dead zones, potentially affecting real-time communication and navigation.

assistance.

10. Estimated Emergency Response Times

- **Rural Areas:** Emergency services may take up to 45 minutes due to limited accessibility and slower response in less populated regions.
- **Urban Areas:** Response times are approximately 20-30 minutes, benefitting from better infrastructure and proximity to emergency services.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Palapatti poses moderate risks, predominantly related to road quality and weather conditions. Concentrated rains can exacerbate road issues, while rural sections may face communication challenges. Safety is generally improved due to low historical incident rates and the availability of alternative routes and emergency services in somewhat reasonable timeframes. Drivers should remain vigilant during peak weather seasons and ensure compliance with transport regulations for safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	High	11.15573, 78.02176	15 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	High	11.11887, 78.01758	15 KM/Hr
12	Turn	High	11.11883, 78.01771	15 KM/Hr
13	Blind Spot	Blind Spot	11.11397, 78.01722	10 KM/Hr
14	Turn	High	11.09256, 78.06435	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.11887, 78.01758



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.11883, 78.01771



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.11397, 78.01722



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.09256, 78.06435
